Final Report – Right to Repair Movement Data Analysis

Caleb Mastruserio

4/24/26

1. Introduction

The Right to Repair Movement has become increasingly important in the modern automotive industry as vehicles grow more complex and dependent on proprietary software systems. In the past, cars were primarily mechanical and could be repaired by independent mechanics or even vehicle owners. However, modern vehicles rely heavily on electronic modules, diagnostic systems, and manufacturer-controlled software, making repairs more difficult and often requiring dealership intervention.

This shift has created a growing issue where consumers are limited in their ability to repair their own vehicles or choose independent repair shops. As a result, repair costs have increased, and consumer choice has been restricted. This project aims to analyze these rising costs and explore how the Right to Repair Movement may help address these challenges.

2. Research Question

How much more expensive are dealership repairs compared to independent shops, and what role can the Right to Repair Movement play in reducing those costs?

3. Data Sources

This analysis uses three primary datasets:

- **CarMD Vehicle Health Index**: Provides data on average vehicle repair costs across different regions and repair types. This dataset helps evaluate overall repair cost trends and differences across vehicles.
- **Consumer Price Index (CPI) for Vehicle Maintenance and Repair**: Sourced from FRED, this dataset shows how vehicle maintenance costs have changed over time, helping identify long-term trends.
- **Car Parts Price Estimation Dataset (Kaggle)**: This dataset provides pricing for common car parts across multiple independent repair shops, allowing for comparison of pricing variability.

4. Methodology

The analysis began with data cleaning and preparation to ensure accuracy and consistency. Several steps were taken:

- Column names were standardized by removing spaces and special characters
- Price columns containing symbols such as “$” were converted into numeric values
- Missing values were identified and removed where appropriate
- Duplicate records were removed to prevent skewing results
- New variables, such as average part price, were created to simplify comparisons

These steps ensured that the datasets were clean and suitable for analysis and visualization.

5. Exploratory Data Analysis (EDA)

Figure 1: Distribution of Repair Costs

A histogram of average repair costs shows that most repairs fall within a moderate price range, while a smaller number of repairs are significantly more expensive. These high-cost outliers suggest the presence of complex or specialized repairs, often associated with modern vehicle systems.

Figure 2: Boxplot of Repair Costs

The boxplot highlights the presence of outliers and variability in repair costs. This indicates that while many repairs are relatively affordable, some repairs can be extremely costly, likely due to advanced electronics or proprietary components.

Figure 3: Average Price of Car Parts Across Shops

The bar chart demonstrates that certain car parts are significantly more expensive than others. This supports the idea that specific components, particularly those tied to modern systems, contribute heavily to rising repair costs.

Figure 4: Price Comparison Across Shops

The line chart shows noticeable variation in prices for the same parts across different independent shops. This suggests that pricing is not standardized and that consumers may benefit from having multiple repair options rather than being restricted to dealerships.

6. Machine Learning

A linear regression model was implemented to predict repair costs based on variables such as car brand, model, and part type. Categorical variables were encoded using one-hot encoding, and the dataset was split into training and testing sets using an 80/20 ratio.

The model demonstrated that repair costs follow predictable patterns based on these features. This suggests that repair pricing is not entirely dependent on proprietary dealership systems and that independent shops can estimate and perform repairs effectively when given access to the necessary information.

7. Results / Key Findings

- Vehicle repair costs have increased significantly over time
- Newer vehicles tend to be more expensive to repair than older ones
- Software complexity and electronic components contribute to higher repair costs
- Independent shop pricing shows consistent and predictable trends
- Repair cost patterns suggest that dealership exclusivity is not strictly necessary

8. Discussion

The findings of this analysis strongly support the concerns raised by the Right to Repair Movement. As vehicles become more technologically advanced, manufacturers have increased control over repair processes through proprietary tools and software. This limits the ability of independent mechanics and consumers to perform repairs, forcing reliance on dealerships.

The data shows that repair costs are not inherently unpredictable or exclusive to dealerships. Independent shops demonstrate consistent pricing patterns and could effectively perform many repairs if given access to the same tools and information. Expanding access through Right to Repair policies could increase competition, reduce costs, and provide consumers with more options.

9. Limitations

There are several limitations to this analysis:

- The datasets used may not represent all vehicle types or repair scenarios

- Some data contained missing or inconsistent values
- Pricing differences across regions and shops may introduce variability
- The analysis does not include all possible repair cost factors, such as labor complexity or regional pricing differences

Despite these limitations, the overall trends observed remain consistent and meaningful.

10. Conclusion

This project demonstrates that vehicle repair costs have increased over time, largely due to the growing complexity of modern vehicles and their reliance on proprietary systems. The analysis shows that independent repair shops are capable of providing consistent and predictable repair pricing, suggesting that dealership exclusivity is not necessary for effective vehicle maintenance.

The Right to Repair Movement offers a potential solution by increasing access to repair tools, software, and information. This could reduce repair costs, increase competition, and restore consumer choice in the automotive repair industry.

11. References

- CarMD Vehicle Health Index: https://carmd.com/pages/vehicle-health-index
- FRED CPI Data: https://fred.stlouisfed.org/series/CUSR0000SETD
- Kaggle Dataset: https://www.kaggle.com/datasets/shorooq77/car-parts-price-estimation